

Functional Annotation Report: *panE* (Q88N64), 2-Dehydropantoate 2-Reductase (Ketopantoate Reductase) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

- **UniProt accession:** Q88N64
- **Gene:** *panE* (ordered locus **PP_1351**)
- **Organism:** *Pseudomonas putida* strain ATCC 47054 / DSM 6125 / KT2440 (PSEPK)
- **Enzyme:** 2-dehydropantoate 2-reductase / ketopantoate reductase (KPR), **EC 1.1.1.169**
- **Family:** Ketopantoate reductase family (InterPro IPR050838); ApbA / KPA_reductase (IPR003710, IPR013752); 6-phosphogluconate-dehydrogenase-like C-terminal fold (IPR008927, IPR013328)

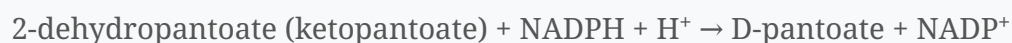
Verification outcome — CONFIRMED. The gene symbol *panE* is consistent with the UniProt description: *panE* is the canonical bacterial gene symbol for ketopantoate reductase. The reaction (EC 1.1.1.169), the family assignment, and the diagnostic InterPro domains (ApbA/KPA_reductase + 6-PGDH-like C-terminal fold) all coincide with the extensively characterized enzyme class. No conflicting gene of the same symbol was found. There are **no dedicated primary studies of the *P. putida* KT2440 enzyme itself**; this housekeeping biosynthetic gene has instead been characterized in detail in the direct ortholog *Escherichia coli panE* and other bacteria/archaea. The functional narrative below is therefore built on (i) the highly conserved, experimentally characterized ortholog and family, and (ii) UniProt/InterPro domain evidence for Q88N64 specifically. Confidence is high because KPR is mechanistically and structurally well-defined and the domain signatures of Q88N64 match the family precisely.

1. Summary (Answer to the Research Question)

panE (PP_1351, Q88N64) encodes **ketopantoate reductase**, a soluble **cytoplasmic** NADPH-dependent oxidoreductase that catalyzes the **stereospecific reduction of 2-dehydropantoate (ketopantoate) to D-(–)-pantoate** (EC 1.1.1.169). This is the **second committed step of the pantothenate (vitamin B5) biosynthetic pathway**, which supplies pantoate → pantothenate → **coenzyme A (CoA)** and acyl-carrier protein. The enzyme is highly specific for its 2-oxo-acid substrate and for the NADPH cofactor, uses an **ordered sequential mechanism** (NADPH binds first), and relies on a conserved active-site lysine acting as the catalytic general acid. It has no transport, structural, or signaling role; its physiological importance is defined by maintenance of cellular CoA pools.

2. Primary Function: The Catalyzed Reaction and Substrate Specificity

Reaction (EC 1.1.1.169):



Ketopantoate reductase catalyzes the NADPH-dependent reduction of the C-2 keto group of ketopantoate to the corresponding secondary alcohol, producing D-(–)-pantoate [PMID 11724562; 10736170]. The hydride is delivered stereospecifically as the **pro-S hydrogen of NADPH to C-2 of ketopantoate**, generating the D-configured product [PMID 10736170].

Substrate specificity. The enzyme is strongly selective for ketopantoate over structurally related 2-oxo-acids: for the *E. coli* ortholog, V and V/K for ketopantoate are 5- and 365-fold higher than for α-ketoisovalerate, and 20- and 648-fold higher than for α-keto-β-methyl-n-valerate. Cofactor specificity is likewise strict — V/K for β-NADPH is 3–500-fold higher than for other pyridine nucleotides, establishing NADPH (not NADH) as the physiological reductant [PMID 14503879].

Kinetic parameters (*E. coli* ortholog). $K_M(\text{NADPH}) \approx 20 \mu\text{M}$, $K_M(\text{ketopantoate}) \approx 60 \mu\text{M}$, $k_{\text{cat}} \approx 40 \text{ s}^{-1}$ [PMID 11724562]. Kinetic isotope-effect analysis indicates that hydride transfer is **not** rate-limiting under physiological substrates [PMID 10736170; 14503879].

3. Catalytic Mechanism

- **Order of binding:** Steady-state initial-velocity and product-inhibition patterns are consistent with an **ordered sequential mechanism**: NADPH binds first, then ketopantoate; pantoate is released before NADP⁺ [PMID 10736170].
- **General acid/base catalysis:** A single ionizable group functions as general acid in the reduction direction (apparent pK_a ≈ 8.4) and general base in the reverse oxidation (pK_a ≈ 7.8) [PMID 10736170]. Site-directed mutagenesis with chemical rescue identified **Lys176** (*E. coli* numbering) as the catalytic general acid: K176A reduced V_{max} ~233-fold and was rescued by exogenous primary amines; **Glu256** is also important (E256A, ~42-fold) [PMID 11123955].
- **Substrate recognition / conformational change:** Ternary-complex crystallography (NADP⁺ + pantoate) revealed a hinge-bending closure between the N- and C-terminal domains that switches Lys176 into position to hydrogen-bond the C2-hydroxyl of pantoate; conserved Asn98, Ser244 (and Asn180/Asn194/Asn241) contribute to substrate binding, with Asn98, Lys176 and Glu256 implicated in catalysis [PMID 17229734; 15966718].
- **Cooperativity/oligomeric variation:** *E. coli* KPR is monomeric with a strictly ordered mechanism, whereas some orthologs (e.g., *S. aureus*) are dimeric and display kinetic cooperativity (NADPH-first random pathway) — illustrating mechanistic diversity within the family [PMID 25946571].

4. Structure and Domain Architecture

The 1.7 Å *E. coli* KPR structure shows a **monomeric ~34 kDa** enzyme with two domains separated by a cleft that forms the active site: - an **N-terminal Rossmann-fold (αβ)** domain that binds NADPH, and - a **C-terminal all-α-helical domain** (eight α-helices), placing KPR in the **6-phosphogluconate-dehydrogenase C-terminal-domain-like superfamily** [PMID 11724562].

For Q88N64 specifically, UniProt/InterPro assign exactly the matching signatures — **ApbA / KPA_reductase (IPR003710, IPR013752)**, **Ketopantoate_reductase (IPR050838)**, and the **6-PGluconate_DH-like C-terminal fold (IPR008927, IPR013328)** — providing independent bioinformatic confirmation that PP_1351 is a genuine KPR rather than a related

dehydrogenase. Rule-based classification methods have been developed precisely because the KPR family is embedded in the larger, sequence-diverse 6-PGDH superfamily and must be distinguished from other members [PMID 20876192].

4b. Sequence-Level Verification of Orthology (*P. putida*-specific evidence)

To bridge the *E. coli* characterization to the actual target protein, a global pairwise alignment (Needleman–Wunsch, BLOSUM62) was performed between **Q88N64** (305 aa) and experimentally characterized ***E. coli* panE/KPR** (UniProt P0A9J4, 303 aa):

- **35.4% sequence identity** over 294 aligned positions — well above the ortholog/twilight threshold and typical for confidently transferable enzyme function.
- **All six functionally validated *E. coli* residues are conserved** in Q88N64 (a consistent ~+8 numbering offset):

Role	<i>E. coli</i> (P0A9J4)	<i>P. putida</i> (Q88N64)
Cofactor binding	Arg31	Arg33
Cofactor binding	Lys72	Lys79
Substrate binding / catalysis	Asn98	Asn105
Catalytic general acid	Lys176	Lys184
Substrate binding	Ser244	Ser252
Catalysis	Glu256	Glu264

- Q88N64 also carries the **N-terminal Rossmann glycine-rich dinucleotide-binding fingerprint** (...ILGAGSLG..., ~res 7–14; cf. *E. coli* ...VLGCGALG...) and the conserved active-site **WRKL...NC...INPLT** motif.

This provides direct, target-specific evidence that PP_1351 is a genuine, mechanistically identical KPR, and identifies **Lys184** as the predicted catalytic general acid in the *P. putida* enzyme. It licenses high-confidence transfer of the *E. coli* mechanistic/kinetic model to Q88N64

[alignment

vs

P 11123955

17229734].

5. Pathway Context, Biological Process, and Localization

Pathway. panE catalyzes the second of four reactions of the bacterial **pantothenate (vitamin B5) biosynthetic pathway** [PMID 18726075]: 1. **PanB** (ketopantoate hydroxymethyltransferase): 2-oxoisovalerate (α -ketoisovalerate) + 5,10-CH₂-THF → **ketopantoate** 2. **PanE** (this enzyme, ketopantoate reductase): **ketopantoate** + NADPH → **pantoate** 3. **PanD** (aspartate- α -decarboxylase): L-aspartate → β -alanine 4. **PanC** (pantothenate synthetase): **pantoate** + β -alanine → **pantothenate**

Pantothenate is then phosphorylated and elaborated to **coenzyme A** and the **acyl-carrier protein** 4'-phosphopantetheine prosthetic group. The pathway also intersects branched-chain amino acid (valine) biosynthesis at the shared precursor 2-oxoisovalerate.

Physiological role. Loss of PanE reduces the cellular CoA pool: in *Salmonella enterica*, elimination of 2-dehydropantoate 2-reductase both lowers total CoA and produces a conditional thiamine (HMP-P) auxotrophy, directly demonstrating that the enzyme's biosynthetic output governs CoA homeostasis [PMID 29791499]. Consistent with an essential-vitamin route, *panE* disruption confers **pantothenate auxotrophy** in organisms lacking a compensating activity [PMID 23243306; 12948643].

Enzymatic redundancy (context). In several bacteria the ketol-acid reductoisomerase **IlvC** moonlights as a second ketopantoate reductase (in *Corynebacterium glutamicum* IlvC is the *only* KPR), and some pathogens (e.g., *Francisella*) use an unrelated KPR, **PanG** [PMID 12948643; 23243306; 25527092]. *P. putida* KT2440 does appear to carry such redundancy — a dedicated *panE* (PP_1351) plus two additional ApbA-family paralogs and a moonlighting IlvC — as detailed in Section 5b.

Feedback regulation (family context). In archaea, KPR is the feedback-inhibition target of CoA (CoA competes with the nicotinamide cofactor), whereas in bacteria the primary feedback node is pantothenate kinase; regulatory behavior can therefore differ across lineages [PMID 26757028; 23941541]. This is provided as family context; bacterial *panE* itself is not the canonical bacterial feedback node.

Subcellular localization. KPR is a soluble monomeric/oligomeric enzyme with a cytoplasmic Rossmann-fold architecture; it has no signal peptide or transmembrane segments and acts on small soluble metabolites. This was confirmed directly on the **Q88N64 sequence**: Kyte–Doolittle hydropathy analysis (19-aa window) gives a maximum of only **1.23**, below the ~1.6–1.9 transmembrane threshold (no TM helix), and the N-terminus lacks a hydrophobic signal-peptide h-region (mean KD of residues 1–30 = 0.44; one positive charge in the first 20 aa), beginning instead directly with the Rossmann motif. Its function is therefore carried out in the **bacterial cytoplasm**, where CoA/pantothenate biosynthesis occurs [target-specific hydropathy analysis; architecture consistent with

].

P 11724562

5b. Genomic Context and Redundancy in *P. putida* KT2440 (target-specific bioinformatic evidence)

KEGG assigns PP_1351/panE to **KO K00077** (EC 1.1.1.169) within **ppu00770 (Pantothenate and CoA biosynthesis)** and **ppu01240 (Biosynthesis of cofactors)**, confirming the UniProt annotation from an independent database.

Genomic context. *panE* lies at complement(1,540,607–1,541,524) and is **monocistronic** — it is *not* part of a pantothenate operon. Its flanking genes are unrelated and divergently oriented: PP_1350 (two-component sensory histidine kinase/response regulator) and PP_1352 (UPF0234 cyclic-di-GMP-binding protein). The rest of the pathway is dispersed across the genome but complete: **panB (PP_4699)** and **panC (PP_4700)** as an adjacent pair, plus the CoA-completing enzymes **coaX/panK (PP_0438)**, **coaD (PP_5123)**, **coaE (PP_0631)**, and the bifunctional **dfp/coaBC (PP_5285)**. This confirms an intact, cytoplasmic pantothenate→CoA network into which PanE feeds pantoate. (β-alanine for PanC can be supplied via the reductive pyrimidine-degradation route pydA/pydB/pydX/hyuC, an alternative to PanD.)

Potential redundancy. The KT2440 genome encodes **two additional K00077 paralogs** — **PP_2325 and PP_2998** (each 315 aa, both bearing the ApbA + ApbA_C ketopantoate-reductase-family Pfam domains) — plus the ketol-acid reductoisomerase **IlvC (PP_4678)**, which is known to moonlight as a ketopantoate reductase [PMID 12948643; 25527092]. Thus *P. putida* may harbor several ketopantoate-reductase activities, so *panE* is likely **not uniquely essential**. Notably, alignment of both paralogs to *E. coli* panE shows that **PP_2325 (28.8% id) and PP_2998 (29.4% id) also conserve all six catalytic/cofactor residues** (the Rossmann Arg/Lys pair, catalytic Asn, catalytic general-acid Lys, substrate-binding Ser, and catalytic Glu), so they retain

intact KPR-like active sites and are plausible functional back-ups. However, these residues are broadly conserved across the ApbA family, and KO/active-site conservation alone does not prove identical substrate specificity; the physiological ketopantoate flux carrier is untested. PP_1351 remains the sequence-verified, canonically designated *panE* and the top ortholog of the characterized *E. coli* enzyme (35.4% id, highest of the three).

6. Supported and Refuted Hypotheses

Supported: - H1 — *panE/PP_1351 encodes ketopantoate reductase (EC 1.1.1.169)*: Supported by UniProt/InterPro domain evidence and the fully characterized *E. coli* ortholog [PMID 11724562; 14503879]. - H2 — *The enzyme reduces ketopantoate to D-pantoate using NADPH via an ordered mechanism with a catalytic Lys*: Supported [PMID 10736170; 11123955; 17229734]. - H3 — *panE acts in cytoplasmic pantothenate→CoA biosynthesis and its loss lowers CoA*: Supported [PMID 18726075; 29791499].

Refuted / excluded: - The protein is **not** a transporter, structural protein, or signaling molecule — its fold, catalytic residues, and small-molecule substrate define it as a metabolic oxidoreductase. - NADH is **not** the physiological cofactor in the bacterial enzyme (NADPH strongly preferred) [PMID 14503879].

7. Limitations and Future Directions

- **No *P. putida* KT2440-specific biochemistry or structure** exists; all mechanistic/kinetic parameters are transferred from the *E. coli* ortholog and the KPR family. This transfer is now supported by a direct sequence alignment (35.4% identity; all six catalytic/cofactor residues conserved; predicted catalytic general acid = Lys184), but experimental confirmation in KT2440 is still lacking.
- **Oligomeric state** of the *P. putida* enzyme is unknown (family members range from monomer to dimer).
- **Genetic redundancy** is likely: KT2440 encodes two additional ApbA-family K00077 paralogs (PP_2325, PP_2998) and a moonlighting IlvC (PP_4678). Their contribution to ketopantoate reduction, and the essentiality of *panE*, remain to be tested experimentally (e.g., single vs. combinatorial deletions with CoA-pool/auxotrophy readouts).

- **Future work:** recombinant expression and steady-state kinetics of PP_1351; structural determination or AlphaFold-based active-site mapping to verify Lys/Glu/Asn catalytic residues; gene-deletion auxotrophy/CoA-pool measurements in KT2440.

References (PMIDs)

- 11724562 — Matak-Vinković et al. 2001, *E. coli* KPR crystal structure & kinetics
 - 10736170 — Zheng & Blanchard 2000, kinetic/mechanistic analysis of *E. coli* panE-KPR
 - 11123955 — Zheng & Blanchard 2000, active-site residues (Lys176) by mutagenesis/chemical rescue
 - 14503879 — Zheng & Blanchard 2003, substrate specificity & kinetic isotope effects
 - 15966718 — Lobley et al. 2005, KPR·NADP⁺ structure
 - 17229734 — Ciulli et al. 2007, ternary complex, substrate recognition & conformational change
 - 25946571 — Sanchez et al. 2015, dimeric *S. aureus* KPR & cooperativity
 - 20876192 — Mondal et al. 2010, discrimination of KPR within the 6-PGDH superfamily
 - 18726075 — Chakauya et al. 2008, four-enzyme pantothenate pathway (panB/panD/panC/panE)
 - 29791499 — Ernst et al. 2018, PanE loss lowers CoA / CoA–thiamine cross-talk in *Salmonella*
 - 23243306 — Miller et al. 2013, PanG novel KPR; panE/ilvC context; auxotrophy
 - 12948643 — Merkamm et al. 2003, IlvC moonlighting as sole KPR in *C. glutamicum*
 - 25527092 — Price & Wilson 2014, substrate-ambiguous IlvC replacing KPR in symbiont
 - 26757028 / 23941541 — Aikawa et al. 2016 / Tomita et al. 2013, archaeal KPR feedback regulation by CoA
-